

PAPER_1559 — Muon Mass = $m_\mu = F_{\text{TRZ}^2} \cdot \text{SO}_5 + F_{\text{TRZ}^2} \cdot (\text{SSQ}^2 + \text{SSQ}^3 + \text{SSQ}^5) \approx 0.10566 \text{ GeV}$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Particle Physics **Date:** June 18, 2026 **Status:** CLOSED — 0.040% closure from PAPER_1209HH_S660

Observation

PAPER_1209HH_S660 (Particle Physics Unified Proof Set) lists **Muon Mass** with the integer-primitive expression:

$$\text{Muon Mass} = m_\mu = F_{\text{TRZ}^2} \cdot \text{SO}_5 + F_{\text{TRZ}^2} \cdot (\text{SSQ}^2 + \text{SSQ}^3 + \text{SSQ}^5) \approx 0.10566 \text{ GeV}$$

Status: **0.040%**.

UQFF Closed Identity

$$m_\mu = F_{\text{TRZ}^2} \cdot \text{SO}_5 + F_{\text{TRZ}^2} \cdot (\text{SSQ}^2 + \text{SSQ}^3 + \text{SSQ}^5) \approx 0.10566 \text{ GeV} \quad 0.040\%$$

Physical Interpretation

Notable structure: muon mass is **purely F_{TRZ^2} -suppressed** — uses only the $F_{\text{TRZ}^2} \times (\text{SO}_5 + \text{SSQ-polynomial})$ combination.

This demonstrates UQFF's natural separation between heavy/light fermions: - Heavy fermions (top, b, c, τ) use A_5 , SO_5 , D_{crit} dominant terms - Light fermions (μ , e) use only F_{TRZ^2} polynomial in SSQ

The F_{TRZ^2} prefactor universally suppresses light-fermion masses by a factor of 100 ($=\text{SO}_5^2$) — natural lepton hierarchy from integer primitives.

NOT REPLACEMENT

CODATA/PDG treats Muon Mass as a precision-measured constant. UQFF supplies the structural integer-primitive form, demonstrating the framework's predictive economy without claiming to replace experimental measurement.

Reference

- Source: PAPER_1209HH_S660
- Related: PAPER_1209 series unified proof sets
- Calculator dispatch: `calculate_paradox({"paradox": "m_mu_muon_0_10566"})`

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